CHAPTER 18

Rate of chemical reactions

Chemical reactions occur at many different rates. The explosion of gunpowder and the combustion of petrol in a car's engine occur very quickly. On the other hand, the chemical weathering of limestone buildings and statues, the ripening of fruit and the rusting of iron all occur quite slowly.

During chemical reactions, particles such as atoms, molecules and ions collide with each other and undergo a rearrangement to produce new substances. It is important to appreciate that collisions between reactant particles do not always result in a chemical reaction. For example, while a car's fuel tank is being filled with petrol, the hydrocarbon molecules in the petrol are colliding with oxygen molecules in the air without a reaction occurring.

By the end of this chapter, you will understand how rates of chemical reactions can be measured. You will also be able to describe how varying the conditions of chemical reactions can affect the rate of a reaction.

Using collision theory, you will learn to predict the effects of concentration of solutions, temperature, surface area, gas pressure and the presence of a catalyst on the rate of chemical reactions, and explain these effects.

Science understanding

- the rate of chemical reactions can be quantified by measuring the rate of formation of products or the depletion of reactants
- varying the conditions under which chemical reactions occur can affect the rate of the reaction
- collision theory can be used to explain and predict the effects of concentration, temperature, pressure, the presence of catalysts and surface area on the rate of chemical reactions
- the activation energy is the minimum energy required for a chemical reaction to occur and is related to the strength and number of the existing chemical bonds; the magnitude of the activation energy influences the rate of a chemical reaction
- energy profile diagrams, which can include the transition state and catalysed and uncatalysed pathways, can be used to represent the enthalpy changes and activation energy associated with a chemical reaction

18.1 Investigating the rate of chemical reactions

FIGURE 18.1.1 How quickly do the chemical reactions involved in baking occur?

The time it takes for a batch of Anzac biscuits to cook in the oven (Figure 18.1.1) and the time taken for a fibreglass patch on a surfboard to set are processes related to the rate of chemical reactions.

In this section, you will learn how changes to reaction conditions affect reaction rates. You will also learn how chemists measure the rate at which a chemical reaction occurs.

FAST AND SLOW REACTIONS

Chemical reactions are taking place all around us:

- in the soil and rocks beneath our feet
- in the air around and above us
- inside every plant and animal
- in our homes, schools and workplaces.

Some of these reactions are over in a flash. In a car accident when a car's airbag needs to be inflated, the chemical reactions producing the gas that expands the airbag need to happen extremely quickly. On the other hand, if the car's painted surface is scratched to expose the metal beneath, the rusting reactions take place at a very slow rate.

FACTORS THAT AFFECT REACTION RATES

Experimental investigations have shown that five main factors can change the rate of a chemical reaction:

- surface area of solid reactants
- concentration of reactants in a solution
- gas pressure
- temperature
- the presence of catalysts.

You can probably think of some examples of situations where one or more of these conditions is changed and a reaction becomes noticeably faster or slower.

FIGURE 18.1.2 Particle size can be used to control the rate of reaction and create different effects during fireworks displays.

Surface area

The surface area of solid reactants can have a significant effect on reaction rate. Smaller particles have a much larger surface area than the same mass of large particles. As a result, the smaller particles react much faster.

Manufacturers of fireworks modify the surface area of solid reactants to control the rate at which fuels in the fireworks burn and create different effects (Figure 18.1.2). For example, very small pieces of aluminium confined in a shell explode violently. If larger pieces of aluminium are used, the reaction is slower and sparks are seen as pieces of burning metal being ejected.

FIGURE 18.1.3 This limestone statue has become pitted in recent years. Limestone (calcium carbonate) reacts more rapidly with the increased concentration of hydrogen ions in rainwater.

Concentration

The concentration of solutes dissolved in a solution can influence the rate of their reactions: higher concentrations usually lead to increased reaction rates.

Pollutants, such as sulfur dioxide and nitrogen dioxide, are released by cars and many industrial processes. When these compounds react with rainwater, acids such as sulfurous acid and nitric acid are formed. This causes the rainwater's hydrogen ion concentration to increase significantly and become **acid rain**. The increasing acidity of rain over the past 200 years has caused many famous marble buildings and statues to deteriorate much more rapidly due to the reaction between marble and the acids (Figure 18.1.3).

CHEMISTRY IN ACTION

Saved by a very fast chemical reaction

Imagine the scene. An 18-year-old borrows his parent's car to take his girlfriend for a drive to celebrate gaining his driver licence. Roof down, enjoying the beautiful afternoon and the countryside, the driver rounds a corner to find the road wet. The car begins to slide on the wet surface. In his inexperience the driver brakes; the car starts to spin. Suddenly, the car is leaving the road and heading straight for a large tree. Then, bang!

Later, the car was estimated to have been travelling at 60 km/h when it hit the tree. The collision was a 'head-on', with the front and passenger side taking most of the impact. Yet the girl in the passenger seat suffered just a chipped tooth, and her boyfriend sustained only minor bruising.

This is the true story of a lucky escape, thanks to a very rapid chemical reaction. As the collision took place, airbags were inflating and then deflating as the travellers were slammed forward towards the windscreen. The driver described it as being 'all over in a flash' and had no clear recollection of the airbags going off.

FIGURE 18.1.4 Airbags are deployed within 30 milliseconds of an impact.

Hidden in the car's steering wheel, dashboard and windscreen pillars, special nylon bags fill with gas within 30 milliseconds of impact (Figure 18.1.4). As a consequence, the car's occupants are prevented from smashing their heads against the steering wheel, dashboard, windscreen or front pillars, all within the blink of an eye. As the head and body strike the airbags, the cushion of gas is forced out of the bag through tiny vents, and within 100 milliseconds the bag has completely deflated.

Air bags contain a mixture of crystalline solids—sodium azide (NaN_3), potassium nitrate (KNO_3) and silica (SiO_2)—stored in a canister. Sensors in the front of the car detect the difference between a bump and life-threatening impact. When a response is required, an electronic impulse initiates a series of three separate reactions. The electronic impulse 'ignites' the sodium azide. Sodium metal and hot nitrogen gas are the products of this energy-releasing, exothermic reaction:

$$2NaN_3(s) \rightarrow 2Na(l) + 3N_2(g)$$

The pulse of hot nitrogen gas released from this reaction starts to inflate the nylon bag. The molten sodium metal immediately reacts with the potassium nitrate, generating more nitrogen gas, as well as sodium oxide and potassium oxide, which are white powdery solids.

The equation for this second reaction is:

$$10Na(l) + 2KNO_3(s) \rightarrow K_2O(s) + 5Na_2O(s) + N_2(g)$$

A filtration system prevents any of the oxides from entering the nylon bag, while a third reaction 'captures' them to produce a harmless glassy solid.

In this third reaction the metal oxides combine with silica:

$$K_2O(s) + Na_2O(s) + SiO_2(s) \rightarrow \text{alkaline silicate ('glass')}$$

Chemical reactions do save lives!

Pressure

In reactions involving gases, increasing the pressure on a reaction increases the rate at which the reaction takes place. Increasing the pressure at constant temperature will result (on average) in the reactant particles becoming closer together.

This will increase the chance of the gas molecules colliding, and therefore increase the rate of reaction. Increasing the pressure of a reacting gas is the same as increasing the concentration because you have the same mass in a smaller volume.

For this reason, engineers often employ high gas pressures in their design of chemical processes that use gas-phase reactions. An example is the production of ammonia gas by reacting hydrogen gas and nitrogen gas. Increasing the pressure ensures a faster rate of reaction.

Temperature

As every cook knows, the temperature of an oven affects the rate of the chemical reactions during baking. The higher the temperature, the more rapidly the reactions occur.

FIGURE 18.1.5 Food is stored at low temperatures in a refrigerator to slow down the rate of reactions that cause food spoilage.

On the other hand, in hot weather it is wise to store fruit and vegetables in the refrigerator so that the chemical reactions that cause them to over-ripen and then spoil will be slowed down at lower temperatures (Figure 18.1.5).

Catalysts

Some chemical reactions occur much more rapidly if another substance is added to the reaction mixture. Such substances are called **catalysts**. Catalysts allow the reaction to follow a more energetically favourable pathway.

For example, if you chew a piece of dry biscuit or bread for several minutes, you may notice it tasting much sweeter. This happens because there is a catalyst present in your saliva that speeds up the breakdown of starch into sweet-tasting sugars.

In the following sections, you will learn how each of these factors can cause these changes in reaction rate.

MEASURING RATES OF REACTION EXPERIMENTALLY

> The rate of reaction is defined as the change in concentration of a reactant or product per unit time.

The **rate of reaction** is defined as the change in concentration of a reactant or product per unit time.

To experimentally determine the rate of reaction, either directly or indirectly, you need to measure how much of a reactant is being used up or how much of a product is being formed in a given time period.

When a reaction involves gaseous products, this might involve measuring changes in mass or gas volume with time.

The graphs shown in Figure 18.1.7 were obtained by measuring the volume of carbon dioxide produced in the reaction between marble chips and hydrochloric acid. The experiment was performed twice, first with large marble chips, then with small marble chips. The rate of the reactions is shown by the gradient of the graphs. The graphs demonstrate two phenomena.

Firstly, the reaction rate of both reactions slows during the reaction, as the reactants are used up. The concentration of the acid will always be highest at the start of the reaction so it makes sense that the initial rate will be the fastest. In both of these examples, the graph levels out, indicating that the reaction that is producing the gas has finished, with one of the reactants being totally consumed.

Secondly, the steeper initial gradient of the graph with small marble chips indicates that the rate of production of carbon dioxide gas is faster with the marble chips that have a higher surface area. Note that in comparing the two situations, in this case the size of the marble chips, a valid comparison can only be made of the initial rate. This is because as soon as the reaction starts, other variables such as concentration of reactants, and possibly temperature will be changing.

Colour changes and pH changes can also be used to follow the rate of some reactions, depending on the colour and acidity of reactants and products.

CHEMFILE

Fireworks and nanoparticles

Studies have shown that the fireworks (Figure 18.1.6) associated with celebrations such as New Year's Eve and the Lantern Festival in China can significantly increase air pollution levels. Sulfur dioxide and particles of metals are released when fireworks explode. These can cause breathing problems and lung disease.

Recent research aimed at lowering the environmental impact of fireworks has focused on reducing the particle size of chemicals used as fuels in the fireworks. By using smaller nanoparticles, reaction rates are increased and smaller amounts of chemicals are needed for the same performance. Thus, fewer pollutants are released into the atmosphere.

However, this new approach carries increased risks because fireworks made of such small particles could be even more explosive.

FIGURE 18.1.6 Fireworks on New Year's Eve, Sydney

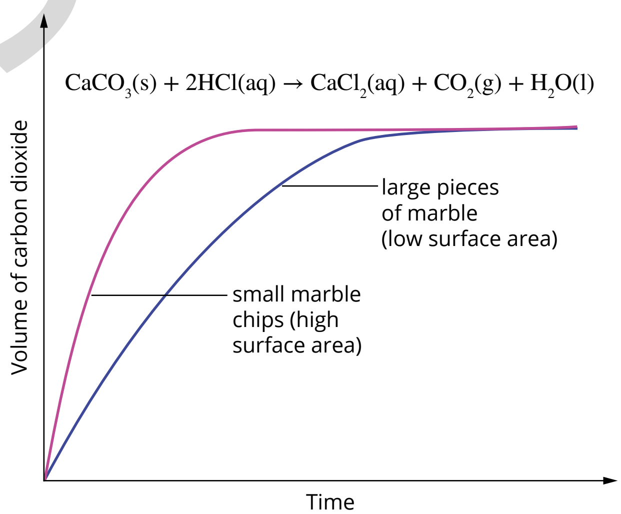


FIGURE 18.1.7 As carbon dioxide is produced from the reaction between marble chips and hydrochloric acid, it is collected in a gas syringe and its volume is recorded.

18.1 Review

SUMMARY

- The rate of a reaction is the change in concentration of reactants or products over time.
- The rate of a reaction may be increased by:
  - increasing surface area
  - increasing the concentration of a reactant
  - increasing the pressure of a gaseous reactant
  - increasing temperature
  - adding a catalyst.
- A range of experimental methods can be used to measure the rate of a reaction, including measuring the following during specific time intervals:
  - mass loss
  - volume of gas produced
  - colour change
  - concentration changes
  - pH change.

KEY QUESTIONS

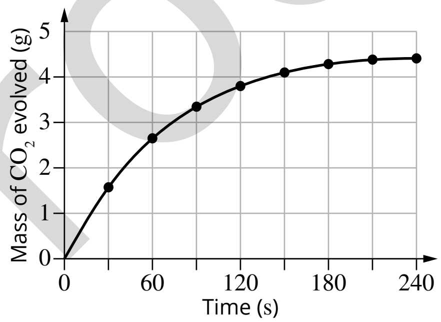
1 Which one of the following changes would *decrease* the rate of the reaction between zinc metal and dilute hydrochloric acid?

A increasing the temperature of the hydrochloric acid
B decreasing the size of the pieces of zinc
C decreasing the concentration of the hydrochloric acid
D decreasing the volume of hydrochloric acid used

2 Select the correct response in the statements about the five main ways in which reaction rates can be increased.

a *increasing/decreasing* surface area of solid reactants
b *increasing/decreasing* the temperature of a reaction mixture
c *increasing/decreasing* the concentration of a reactant in solution
d *increasing/decreasing* the pressure of gaseous reactants
e *adding/removing* a catalyst

3 The following graph shows the mass of carbon dioxide gas produced during a 4-minute period of a reaction between marble chips (calcium carbonate) and 1.0 mol L^{-1} nitric acid.



a Write a full chemical equation for this reaction.
b Explain whether the rate of this chemical reaction is increasing or decreasing over time.

4 Describe one way of increasing the rate of each of the following reactions.

a wood burning on a camp fire
b removing excess mortar from between bricks using brick cleaner
c baking a cake in the oven

18.2 Collision theory

The chemical equation for a reaction shows the nature of the reactants and products. It provides no information about the way in which the reaction proceeds.

Look at the equation for the reaction of zinc metal with hydrochloric acid:

$$2HCl(aq) + Zn(s) \rightarrow ZnCl_2(aq) + H_2(g)$$

This equation gives no indication about whether the reaction proceeds quickly or slowly; nor can you tell how the products have been formed.

In fact, the rate of this reaction can vary dramatically by changing the size of the zinc pieces or the concentration of acid used. If concentrated acid is used for example, you see vigorous bubbling as hydrogen is rapidly produced. When you reduce the concentration of the acid the bubbling slows as the reaction occurs more slowly (Figure 18.2.1).

Chemical reactions occur as a result of collisions between the reacting particles. This idea is part of the **collision theory** of reaction rates, which will be discussed in this section.

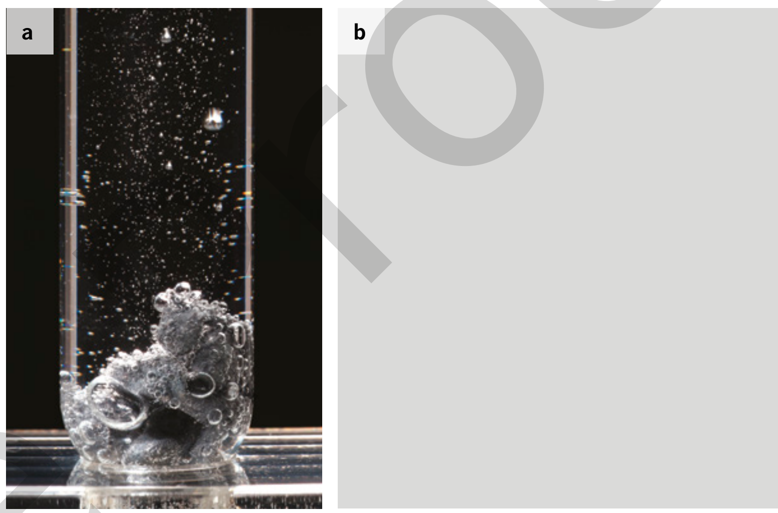


FIGURE 18.2.1 The reaction of zinc metal in a (a) dilute and a (b) concentrated solution of hydrochloric acid

COLLISION THEORY AND ACTIVATION ENERGY

During chemical reactions, particles (atoms, molecules or ions) collide and are rearranged to produce new particles. Consider the decomposition reaction of hydrogen peroxide:

$$2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$$

The collision that forms the first step of the reaction occurs between the two hydrogen peroxide molecules. If this collision is to result in the formation of molecules of water and oxygen, the collision must occur in such a way that the covalent bonds in the hydrogen peroxide break. To break bonds, energy is required.

The collision theory of reactions explains why some collisions result in reactions and others do not. According to collision theory, for a reaction to occur, the reactant particles must:

- collide with each other
- collide with sufficient energy to break the bonds within the reactants
- collide with the correct orientation to break the bonds within the reactants and so allow the formation of new products.

If a collision does not meet all of these requirements, then no reaction occurs. In fact, most collisions do not result in a chemical reaction. Collision theory helps us to explain why this is the case.

Activation energy

For a reaction to occur between reactant molecules, the molecules must collide with a certain minimum amount of energy. Unless this minimum amount of energy is met or exceeded, the colliding molecules will rebound and simply move away from each other without reacting.

The minimum energy that a collision must possess for a reaction to occur is called the activation energy, E_a. When the energy of a collision is equal to or greater than the activation energy, a reaction can occur.

Activation energy can be represented on an energy profile diagram. An energy profile diagram represents the enthalpy (chemical potential energy) of the reactants and the products over the course of the reaction.

Energy profile diagrams for both exothermic (Figure 18.2.2) and endothermic (Figure 18.2.3) reactions have a peak that represents the activation energy. This is sometimes referred to as the activation energy barrier, and represents the minimum energy that must be absorbed in order to break the bonds of reactants so that a chemical reaction can progress. The activation energy is measured from the enthalpy of the reactants to the top of the peak.

You will recall from chapter 10 that an **exothermic reaction** releases more heat energy during the reaction than it absorbs. An **endothermic** reaction absorbs more heat energy during the reaction than it releases. This is represented on the diagram as ΔH and is the difference in enthalpy between the reactants and the products (Figures 18.2.2 and 18.2.3).

Reactant particles must have energy equal to or greater than the activation energy before a reaction can occur.

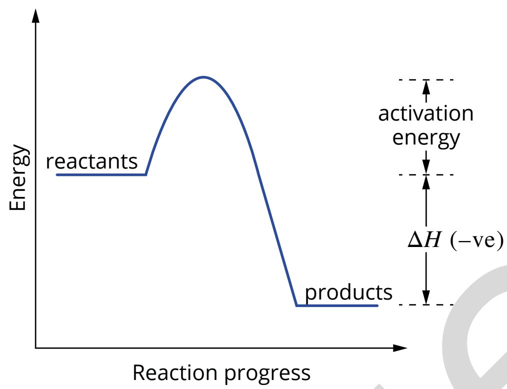


FIGURE 18.2.2 Energy profile diagram for an exothermic reaction showing E_a and ΔH. The ΔH value is negative as overall energy is released during the reaction.

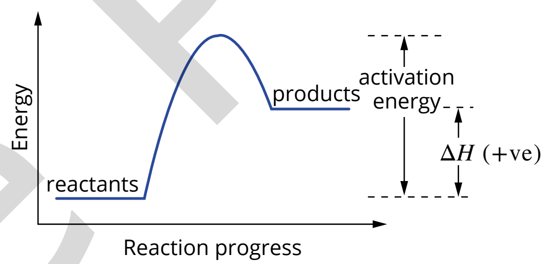


FIGURE 18.2.3 Energy profile diagram for an endothermic reaction showing E_a and ΔH. The ΔH value is positive as overall energy is absorbed during the reaction.

Transition state

When the activation energy is absorbed, a new arrangement of the atoms known as the **transition state** occurs. The transition state occurs at the stage of maximum potential energy in the reaction: the activation energy (Figure 18.2.4). Bond breaking and bond forming are both occurring at this stage, and the arrangement of atoms is unstable. The atoms in the transition state rearrange into the products as the reaction progresses.

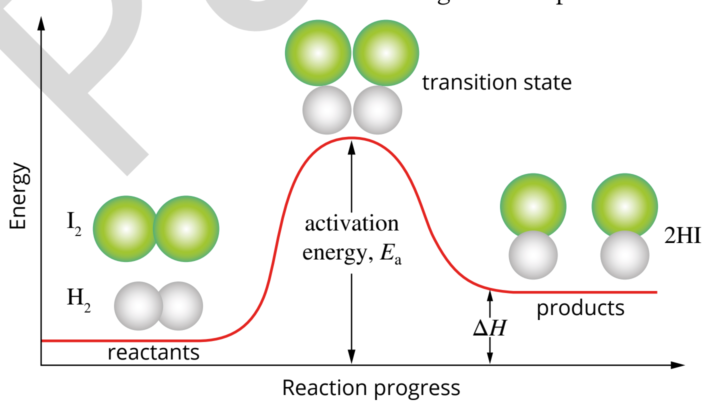


FIGURE 18.2.4 Energy profile diagram for the endothermic reaction $H_2(g) + I_2(g) \rightarrow 2HI(g)$

Activation energy and reaction rate

The magnitude of the activation energy determines how easy it is for a reaction to occur and therefore what proportion of collisions results in a successful reaction. For this reason, the reaction rate is dependent upon the activation energy.

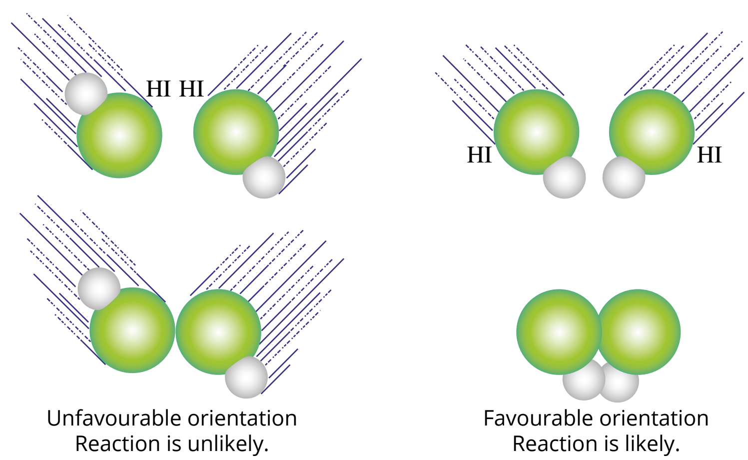
The existence of an activation energy for a reaction means that collisions between reactants do not always result in a chemical change. For example, nitrogen (N_2) and oxygen (O_2) molecules collide frequently in the air around us at room temperature. However, it is only when energy is provided by a spark, such as in car engines or during a lightning strike, that the energy of the collisions is increased enough to overcome the activation energy barrier. This allows nitrogen monoxide to be produced. The nitrogen monoxide formed in this process can then react to form brown nitrogen dioxide (NO_2), a poisonous gas that is a major contributor to the formation of the **photochemical smog** seen in Figure 18.2.5.

FIGURE 18.2.5 Photochemical smog, such as seen here over Barcelona, Spain, in December 2013, is caused by NO_2, a poisonous gas.

Orientation of colliding molecules

For a reaction to occur, reactants need to collide with enough energy to provide the activation energy. Reacting molecules must also collide with each other in the correct orientation in such a way that particular bonds in the reactants are broken and new bonds are formed in the products.

Figure 18.2.6 shows the importance of collision orientation. In the decomposition of hydrogen iodide gas into hydrogen gas and iodine gas, two hydrogen iodide molecules must collide with hydrogen and iodine atoms orientated towards each other for a reaction to possibly occur. If the collision orientation is incorrect, the particles simply bounce off each other, and no reaction occurs.



$$2HI(g) \rightarrow H_2(g) + I_2(g)$$

FIGURE 18.2.6 A reaction between colliding molecules is more likely to occur if the orientation of the collision is favourable.

18.2 Review

SUMMARY

- The activation energy of a reaction is the minimum amount of energy required to break reactant bonds to allow a reaction to proceed.
- The transition state is an unstable state in which bonds in the reactants are being broken and bonds in the products are starting to form.
- An energy profile diagram shows the activation energy as the difference between the enthalpy of the reactants and the maximum energy reached.
- Collision theory is a theoretical model that explains rates of chemical reactions in terms of collisions between particles during a chemical reaction.
- According to collision theory, for a reaction to occur, the reactant particles must:
  - collide with each other
  - collide with sufficient energy
  - collide with the correct orientation.

KEY QUESTIONS

1 According to the collision theory, which one of the following is *not* essential for a reaction to occur?

A Molecules must collide to react.
B The reactant particles should collide with the correct orientation.
C The reactant particles should collide with enough energy to overcome the activation energy barrier.
D The reactant particles should collide with double the energy of the activation energy.

2 Which one of the following is the energy required to produce the transition state in a reaction?

A activation energy
B difference in energy between the products and reactants
C difference in energy between the products and the activation energy
D transition state energy

3 When 1 mol of methane gas burns completely in oxygen, the process of bond breaking uses 3380 kJ of energy and 4270 kJ of energy is released as new bonds form.

a Write a balanced chemical equation for the reaction.
b Calculate the value of the heat of reaction, ΔH, for the reaction.
c Draw and label a diagram to show the changes in energy during the course of the reaction.

4 The formation of hydrogen iodide from its elements is represented by the equation:

$H_2(g) + I_2(g) \rightarrow 2HI(g)$

This reaction has an activation energy of 167 kJ mol^{-1} and the heat of reaction, ΔH, is +28 kJ mol^{-1}.

a Draw and label an energy profile diagram for the reaction.
b Is the reaction as described, the forward reaction, endothermic or exothermic? Explain.

Consider the reverse reaction.

c What is the ΔH for the reverse reaction?
d Calculate the activation energy for the reverse reaction, the decomposition of 2 mol of hydrogen iodide.

18.3 Applying collision theory

Earlier in this chapter you saw that the rate of a reaction can be changed by the:

- surface area of a solid reactant
- concentration of reactants in a solution
- pressure of any gaseous reactants
- temperature of the reaction
- presence of a catalyst.

In any given reaction mixture, only a certain proportion of the collisions are successful. To increase a reaction rate, you can increase the proportion of successful collisions by:

- increasing the frequency of successful collisions by increasing the number of collisions that can occur in a given time
- increasing the proportion of collisions with an energy equal to or greater than the activation energy.

In this section, you will consider how various changes to conditions can affect the proportion of successful collisions that occur between reactant particles and hence change the rate of reaction.

INCREASING THE FREQUENCY OF COLLISIONS

The rate of a reaction increases as the frequency of collisions increases. The frequency of collisions between reactants can be increased by:

- increasing the concentration of the reactants. Collisions occur more frequently when particles are closer together
- increasing the surface area of a solid reactant.

Changing concentration or pressure

In section 18.1, you learnt that the rate of a reaction can be increased by increasing the concentration of a reacting solution or the pressure of a reacting gas. This can be explained by collision theory.

The rate of a reaction increases when the frequency of collisions between reactants increases. When the concentration of a solution increases, there are more reactant particles moving randomly in a given volume of solution (Figure 18.3.1). More particles increases the frequency of collisions, which increases the number of successful collisions in a given time.

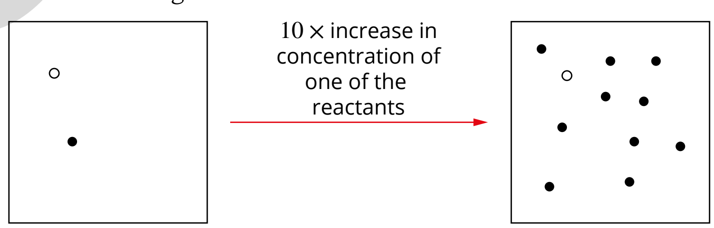


FIGURE 18.3.1 On the left, the concentration of both reactants is low. On the right, the concentration of one of the reactants has been increased ten-fold, resulting in an increase in collision frequency.

For a reaction in the gas phase, the pressure of the gases can be increased either by adding more gas to a fixed volume container or by decreasing the volume of a container with a variable volume, such as a gas syringe. Increasing the pressure increases the concentration of gas molecules, causing more frequent collisions and increasing the number of successful collisions in a given time.

Changing surface area

When a solid is involved in a reaction, only the particles at the surface of the solid are involved in the reaction. The number of particles at the surface depends on the **surface area** of the substance. As you can see in Figure 18.3.2, breaking a solid into smaller parts means that more particles are present at the surface and available to react. The surface area has increased.

As a consequence of the greater number of exposed particles, the frequency of collisions between these particles and the other reactant particles increases, and so the reaction occurs more rapidly.

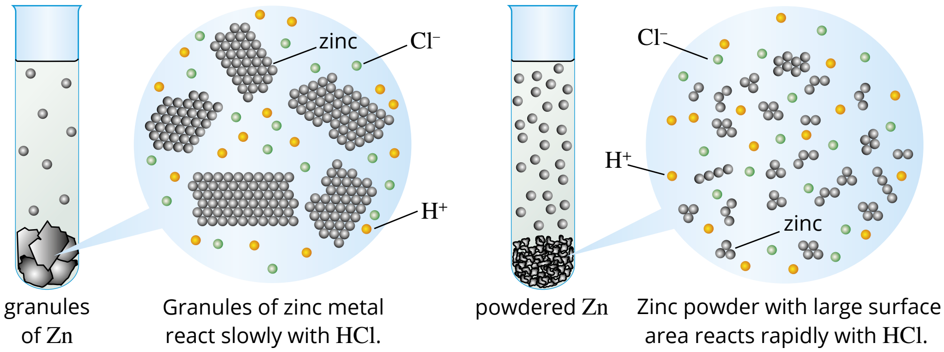


FIGURE 18.3.2 The reaction of hydrochloric acid and zinc. As the surface area of zinc increases, the rate of reaction with hydrochloric acid increases. (Hydrochloric acid exists as Cl^- ions and H^+ ions in solution.)

The effect of increasing surface area can be seen when setting up an open fire at home or on a camp. It is best to first light a pile of kindling rather than trying to directly light large logs. The kindling has a larger surface area than the logs, so it catches fire more easily and will then burn rapidly, providing enough sustained heat energy to make the logs catch fire as well.

When using collision theory to explain the effect of concentration, pressure and surface area on the rate of a chemical reaction, you must discuss the effect on either collision frequency or the number of successful collisions per unit time.

Worked example 18.3.1

USING COLLISION THEORY TO EXPLAIN CHANGES IN RATES OF REACTIONS

The most common fuel sources for explosions in underground mines are flammable gases and explosive dust. Explain, in terms of collision theory, why the presence of coal dust can cause explosions in underground coal mines.	
Thinking	**Working**
Note that this answer uses an explanation structure called the Premise, Reasoning, Outcome (PRO) approach. The premise is the knowledge, theory or the law that is relevant to the situation. The reasoning is where you apply this knowledge or understanding to the situation, and the outcome is the observed result. You can try to use this approach in the Try yourself example.	
Premise: What is the key knowledge, law or theory that you are using?	The more successful collisions the greater the rate of reaction.
Reasoning: Consider the state of the reactants. Relate the state of the reactant to the factor that affects the reaction rate and explain in terms of collision theory.	Coal is a solid. In the mine, there would be lumps of coal and also powdered coal. The surface area of powdered coal is greater than that of solid coal. When the surface area increases, the frequency of collisions increases and so the rate of reaction increases.
Outcome: Return to the question to complete your answer.	The very large surface area of the coal dust allows for an increase in the frequency of collisions with reacting particles, which increases the reaction rate so much that explosions occur.

Worked example: Try yourself 18.3.1

USING COLLISION THEORY TO EXPLAIN CHANGES IN RATES OF REACTIONS

Iron anchors recovered from shipwrecks at considerable depths can show little corrosion after years in the sea, whereas anchors recovered from shallow water are badly corroded. Explain this observation in terms of collision theory.

INCREASING THE ENERGY OF COLLISIONS

As you have seen, a reaction can be made to occur more rapidly by increasing the concentration of the reactants or, for solids, increasing the surface area. A change in temperature can also have a major effect on the rate of a reaction. An increase in temperature not only increases the frequency of collisions, it also increases the kinetic energy of the particles and hence the energy of their collisions.

Maxwell–Boltzmann distribution

At any particular temperature, the particles in a substance have a range of kinetic energies. Although most of the particles have similar kinetic energies, there are always some particles with a high energy or a low energy. This range of energies is shown on a graph called a **Maxwell–Boltzmann distribution curve,** also known as a kinetic energy distribution diagram. Figure 18.3.3 shows how the distribution of energies is represented in a Maxwell–Boltzmann distribution curve.

The vertical axis gives the number of particles with a particular energy, while the horizontal axis gives the kinetic energies of the particles. The area under the curve represents the sum of all the reactant particles. So a **kinetic energy distribution diagram** considers all reacting particles and represents their probable energy distribution, while an energy profile diagram represents the energy of individual particles during the course of a reaction.

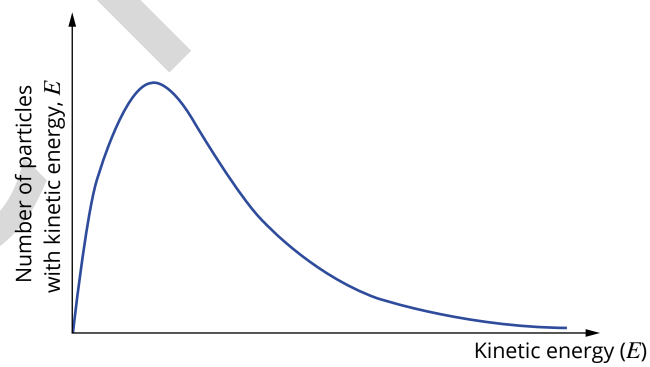


FIGURE 18.3.3 This Maxwell–Boltzmann curve shows the distribution of energies of particles in a sample at a particular temperature. The peak of the graph corresponds to the energy of the greatest number of particles.

During a reaction at a given temperature, only a small proportion of the reactant particles have kinetic energy that is equal to or greater than the activation energy and so are able to react. You can see this in Figure 18.3.4 as a small shaded area to the right of the activation energy, E_a.

Effect of temperature and rate of reaction

The relationship between kinetic energy and velocity is given by the formula $KE = \frac{1}{2}mv^2$. As the temperature of a reaction system increases, the average kinetic energy of the particles increases and the average speed of the particles in the system increases as well.

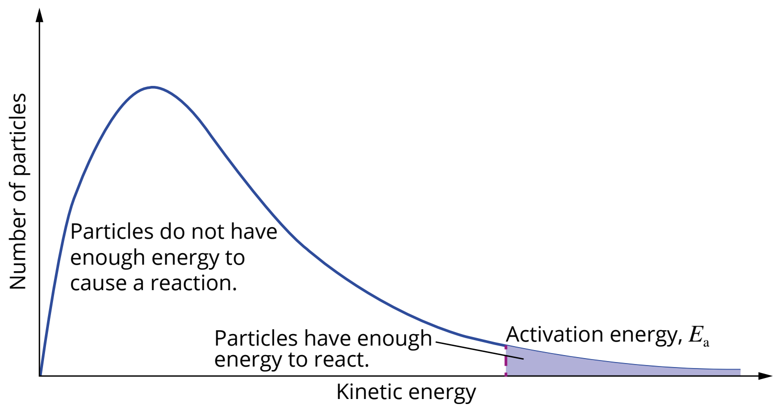


FIGURE 18.3.4 Only a small number of higher energy particles (represented by the shaded area) have sufficient energy to overcome the activation energy barrier.

This is illustrated in Figure 18.3.5 in which the range of kinetic energies for a gas at three different temperatures is shown. Note that the area under the curve, which is equal to the total number of particles in the sample, stays constant when the temperature is changed. As the temperature increases, the increasing average kinetic energy of the particles can be seen by the movement to the right of the peak in the Maxwell–Boltzmann curve.

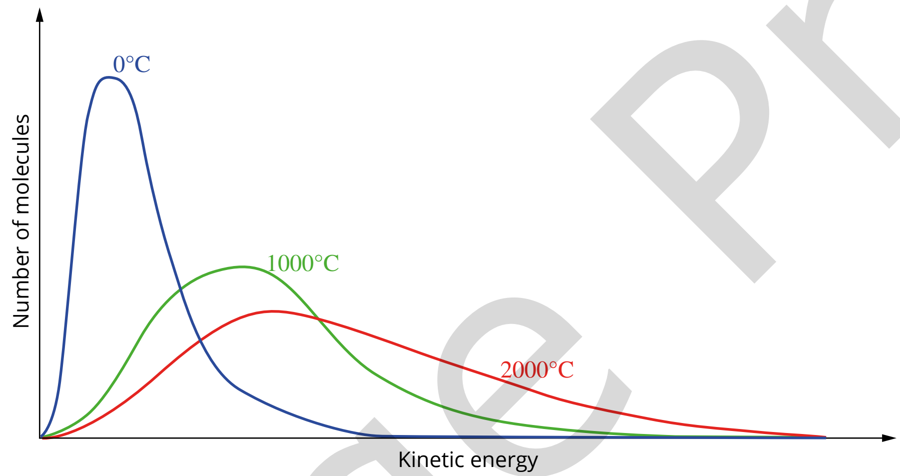


FIGURE 18.3.5 Kinetic energy distribution for a gas at a range of temperatures

As the temperature of a reaction increases, the increased speed of particles causes more collisions, increasing the frequency of successful collisions and the rate of reaction.

The increased kinetic energy of the particles also means that collisions occurring at higher temperatures have greater energy than those at lower temperatures. More particles will have energies that are greater than or equal to the activation energy (Figure 18.3.6) and so the proportion of successful collisions also increases.

When the temperature increases, the increase in reaction rate due to the increased energy of the collisions significantly outweighs the increase in reaction rate due to the increased frequency of collisions.

A temperature increase of just 10°C causes the rate of many reactions to double, but it can be shown that this is not just due to the increased frequency of collisions. The frequency of collisions only increases by about 3% when the temperature increases by 10°C. The main reason why the reaction rate increases is that more particles have sufficient energy to overcome the activation energy barrier of the reaction.

The effect of increasing temperature on the rate of reaction is mainly through increasing the proportion of reacting particles that have energies equal to or greater than the activation energy for the reaction. This increases the proportion of successful collisions.

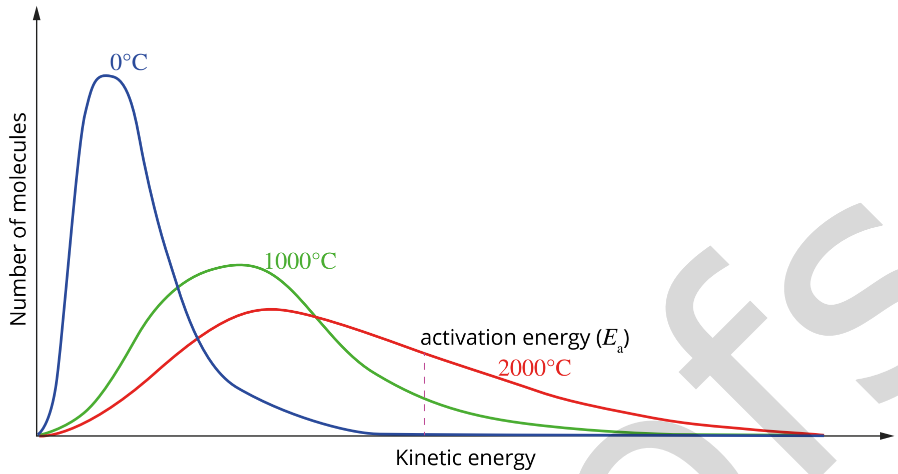


FIGURE 18.3.6 Kinetic energy distribution for a gas at a range of temperatures showing the activation energy for the combustion of this gas. The higher energy particles (those to the right of the E_a) have enough energy to react. Note that there are more particles at the higher temperatures with enough energy to react. This is shown by an increase in the area under the section of the curve above the activation energy as the temperature increases.

FIGURE 18.3.7 The decomposition of hydrogen peroxide is usually very slow, but the addition of crystals of potassium permanganate results in the rapid evolution of oxygen gas and water vapour.

Effect of a catalyst on rate of reaction

Look at the equation for the decomposition of hydrogen peroxide:

$$2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$$

This reaction normally occurs very slowly. When a catalyst, such as crystals of potassium permanganate, are added to the hydrogen peroxide, the reaction occurs rapidly, producing so much oxygen gas and heat that the reaction mixture foams and some of the liquid water vaporises (Figure 18.3.7).

The change in reaction rate when a catalyst is present is often very substantial. The action of a catalyst can be understood using collision theory and the changes in energy that occur during a chemical reaction. Catalysts speed up a reaction by reducing the activation energy barrier for the reaction. You will study the effects of catalysts in more detail in Chapter 19.

Catalysts lower the activation energy for a reaction.

CHEMFILE

Ötzi the Iceman

In September 1991, Erika and Helmut Simon were walking in the Ötztal Alps near the border between Austria and Italy when they discovered the body of what they thought was a dead mountaineer (Figure 18.3.8). It was known that, in this region, bodies decompose very slowly because they are enclosed in ice. Closer examination of the body, and the Bronze Age items with it, eventually established that he had died approximately 5300 years ago.

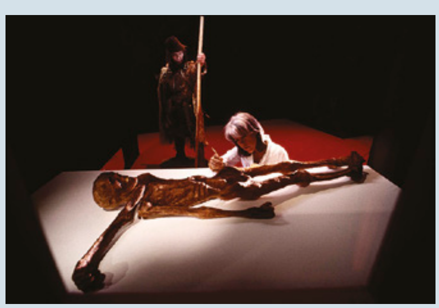


FIGURE 18.3.8 Ötzi the Iceman was so well preserved for 5300 years in the ice that his stomach contents could be identified and pollen of a spring plant was found on his clothes.

18.3 Review

SUMMARY

- Collision theory can be used to explain the increase in rate of reaction by:
  - an increase in concentration of a reactant solution
  - an increase in pressure of a gaseous reactant
  - an increase in surface area of a solid reactant
  - an increase in temperature
  - use of a catalyst.
- Increase in concentration, pressure or surface area results in an increase in the:
  - frequency of collisions between reactants
  - number of successful collisions in a given time.
- Increase in temperature results in an increase in the:
  - frequency of collisions between reactants
  - number of successful collisions in a given time
  - energy of collisions, so an increased proportion of collisions has an energy larger than the activation energy for the reaction.
- A catalyst speeds up a reaction by
  - reducing the activation energy barrier for the reaction

KEY QUESTIONS

1 Which one of the following alternatives correctly explains why a sample of magnesium reacts more rapidly with 1 M HCl than with 0.1 mol L^{-1} HCl?

A The energy of collisions between reactant particles is greater for the reaction containing 1 mol L^{-1} HCl.

B The rate of collisions between reactant particles is greater for the reaction containing 0.1 mol L^{-1} HCl.

C There are more collisions between the magnesium and 1 mol L^{-1} hydrochloric acid.

D The frequency of collisions between reactant particles is greater for the reaction containing 1 mol L^{-1} HCl.

2 Which one or more of the following may be true if a reaction is observed to proceed very slowly?

A The activation energy may be very large.

B The temperature may be low.

C Few collisions may be occurring with the correct orientation.

3 A number of experiments involving the reaction between 100 mL of hydrochloric acid and 5 g of calcium carbonate were carried out. Rearrange experiments A–F to place them in increasing order of rate of reaction (slowest to fastest).

A Powdered $CaCO_3$ and 1 mol L^{-1} HCl are mixed at 40°C.

B Small pieces of $CaCO_3$ and 1 mol L^{-1} HCl are mixed at 15°C.

C Powdered $CaCO_3$ and 1 mol L^{-1} HCl are mixed at 25°C.

D Large pieces of $CaCO_3$ and 0.5 mol L^{-1} HCl are mixed at 15°C.

E Powdered $CaCO_3$ and 1 mol L^{-1} HCl are mixed at 15°C.

F Small pieces of $CaCO_3$ and 0.5 mol L^{-1} HCl are mixed at 15°C.

4 Account for the following observations with reference to the collision model of particle behaviour.

a Surfboard manufacturers find that fibreglass plastics set within hours in summer but may remain tacky for days in winter.

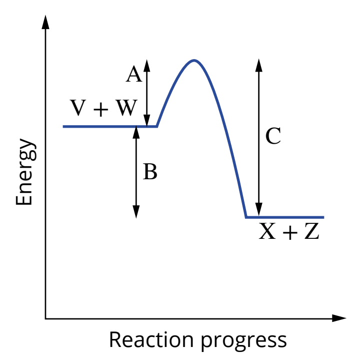
b A bottle of fine aluminium powder has a caution sticker warning ‘Highly flammable, dust explosion possible’.

c A potato cooks much more slowly in a pot of boiling water on a trekking holiday in Nepal than a potato boiled in a similar way in the Australian bush.

5 Consider the reaction between solutions V and W that produces X and Z according to the equation:

$V(aq) + W(aq) \rightarrow X(aq) + Z(aq)$

The energy profile diagram for this process is shown below.



a What do A, B and C represent?

b What effect would the use of a catalyst have on the values of A, B and C?

Chapter review

18

KEY TERMS

acid rain
catalyst
collision theory
photochemical smog
rate of reaction
surface area
transition state

Investigating the rate of chemical reactions

1 Which one of the following is the correct definition of rate of reaction?

A the time it takes for all of a reactant to be used up
B how fast a reaction is going at the end of 1 minute
C how much a reaction is bubbling
D the change in concentration of reactants or products over time

2 Which of the following combinations of reactants will produce the greatest initial reaction rate?

$2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

A $CaCO_3$ chips and 1 mol L^{-1} HCl
B $CaCO_3$ chips and 2 mol L^{-1} HCl
C $CaCO_3$ powder and 2 mol L^{-1} HCl
D $CaCO_3$ powder and 1 mol L^{-1} HCl

3 The following changes are made to a reaction mixture. Which change will lead to a decrease in reaction rate?

A Smaller solid particles are used.
B The temperature is decreased.
C A catalyst is added.
D The concentration of an aqueous reactant is increased.

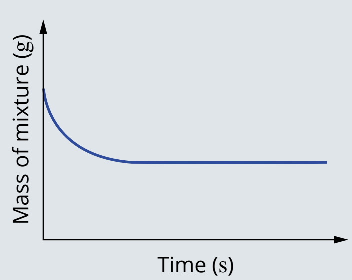
4 Which of the following is the correct unit for measuring the rate of a reaction?

A $mol^{-1}\,L\,s^{-1}$
B $mol\,L^{-1}\,s^{-1}$
C $mol^{-1}\,L^{-1}\,s$
D $mol\,L\,s^{-1}$

5 A 5.00 g piece of copper was dissolved in a beaker containing 500 mL of 2.00 mol L^{-1} nitric acid. The equation for the reaction that occurred is:

$3Cu(s) + 8HNO_3(aq) \rightarrow 3Cu(NO_3)_2(aq) + 2NO(g) + 4H_2O(l)$

The changing mass of the mixture was observed for a period of time, and the following graph was obtained.



a Explain why the mass of the mixture initially decreases with increasing time.

b Based on the information provided, determine which reactant is limiting.

c Redraw the graph, then sketch in the expected curve if 500 mL of 1.00 mol L^{-1} nitric acid had been used instead. Label your new graph line. Explain the difference in shape.

d Redraw the graph, then sketch in the expected curve if 5.00 g of powdered copper was used instead. Label this new graph line. Explain the difference in shape.

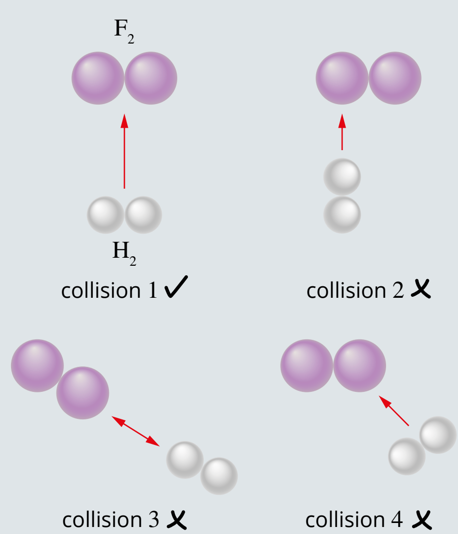
Collision theory

6 According to collision theory, what must happen for a reaction to occur?

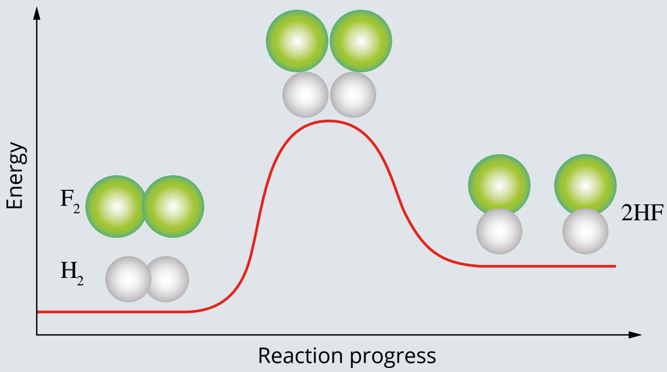
7 The following figure shows the reaction between hydrogen gas and fluorine gas to make hydrogen fluoride. The equation for the reaction is:

$H_2(g) + F_2(g) \rightarrow 2HF(g)$

Using collision theory, explain why collision 1 might be successful while collisions 2–4 will not be successful.



8 The figure below is an energy profile diagram for the substitution reaction between fluorine gas and hydrogen gas.



a Copy this diagram and label ΔH and activation energy.

b Explain what is meant by the term ‘activation energy’.

c Is the reaction endothermic or exothermic?

d Label the transition state in this reaction on your diagram.

e What bonds are beginning to be broken and formed to produce the transition state?

9 Hydrogen reacts explosively with oxygen to form water.

a What chemical bonds are broken in the reaction?

b What chemical bonds are formed?

c Explain how the energy changes during bond-breaking and bond-forming affect the overall energy change for the reaction.

d Why is there no reaction until the reaction mixture is ignited?

Applying collision theory

10 Which one of the following alternatives correctly explains why the rate of reaction between 1 mol L^{-1} $CuSO_4$ and powdered zinc is greater than with an equal amount of large zinc pieces.

A The energy of collisions between the Cu^{2+} ions and powdered zinc is greater than with the large zinc pieces.

B The frequency of collisions between the Cu^{2+} ions and powdered zinc is greater than with the large zinc pieces.

C The energy of collisions between the Cu^{2+} ions and large zinc pieces is greater than with the powdered zinc.

D The frequency of collisions between the Cu^{2+} ions and large zinc pieces is greater than with the powdered zinc.

11 Which one of the following statements correctly describes what must occur when reactant particles collide and react?

A Colliding particles must have an equal amount of kinetic energy.

B Colliding particles must have different amounts of kinetic energy.

C Colliding particles must have kinetic energy equal to or greater than the average kinetic energy.

D Colliding particles must have kinetic energy equal to or greater than the activation energy of the reaction.

12 Account for the following observations with reference to the collision model of particle behaviour.

a Refrigeration slows down the browning of sliced apples.

b Hydrogen gas burns in air to produce water vapour. Using pure oxygen gas instead of air increases the rate of this reaction.

13 Which one of the following factors would *not* increase the rate of decomposition of hydrogen peroxide?

$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$

A increasing the pressure of oxygen gas

B increasing the concentration of hydrogen peroxide

C increasing the temperature of hydrogen peroxide

D adding a potassium iodide catalyst

14 Which pair of statements is correct for the effects of catalyst and concentration on the rate of reaction?

	Adding a catalyst	Increasing the concentration
A	Collision frequency increases	Collision frequency increases
B	Activation energy decreases	Activation energy decreases
C	Activation energy decreases	Collision frequency increases
D	Collision frequency increases	Activation energy decreases

15 a What are the five factors that influence the rate of a reaction?

b Classify the five factors from part **a** according to whether they increase the proportion of successful collisions by:

i increasing collision frequency

ii increasing the proportion of collisions that have energy equal to or greater than the activation energy.

Connecting the main ideas

16 Lumps of limestone, calcium carbonate, react readily with dilute hydrochloric acid. Four large lumps of limestone, mass 10.0 g, were reacted with 100 mL 0.100 mol L^{-1} acid.

- **a** Write a balanced equation to describe the reaction.
- **b** Which reactant is in excess? Use a calculation to support your answer.
- **c** Describe a technique that you could use in a school laboratory to measure the rate of the reaction.
- **d** 10.0 g of small lumps of limestone will react at a different rate from four large lumps. Will the rate of reaction with the smaller lumps be faster or slower? Explain your answer in terms of collision theory.
- **e** List two other ways in which the rate of this reaction can be altered. Explain your answer in terms of collision theory.

17 Chemical reactions in the body normally take place at 37°C. Explain how the rate of chemical reactions in the body can account for the following facts.

- **a** The body often responds to illness by an increase in temperature, accompanied by a higher pulse rate and faster breathing.
- **b** People rescued from drowning after 20–30 minutes in freezing water can sometimes survive and recover with no brain damage.

18 The first step in most toffee recipes is to dissolve about 3 cups of sugar in 1 cup of water. Although sugar is quite soluble in water, this step could be time-consuming. Use your knowledge of reaction rates to suggest at least three things you could do to increase the rate of dissolution without ruining the toffee.